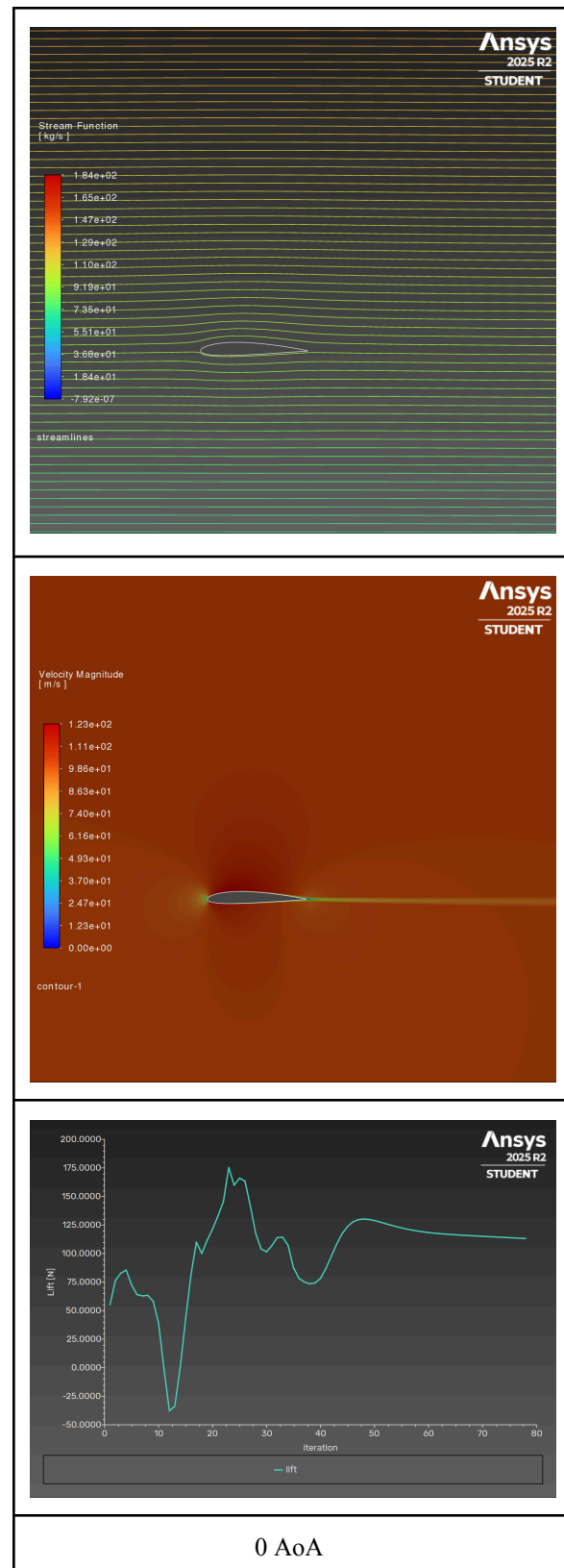


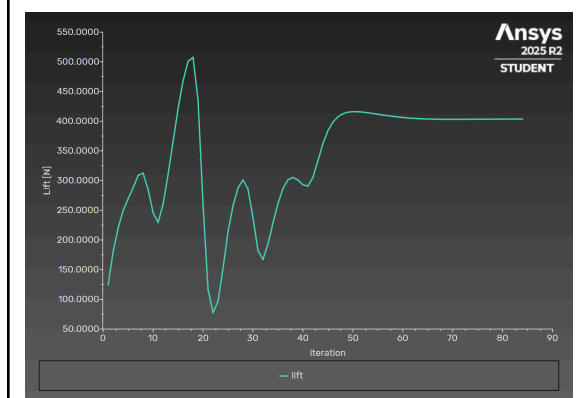
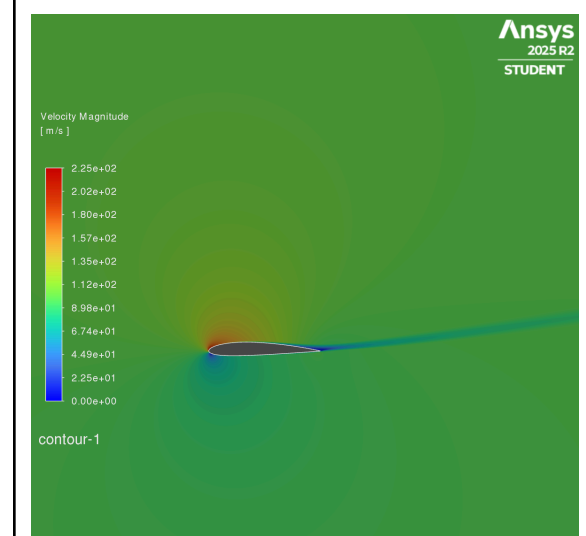
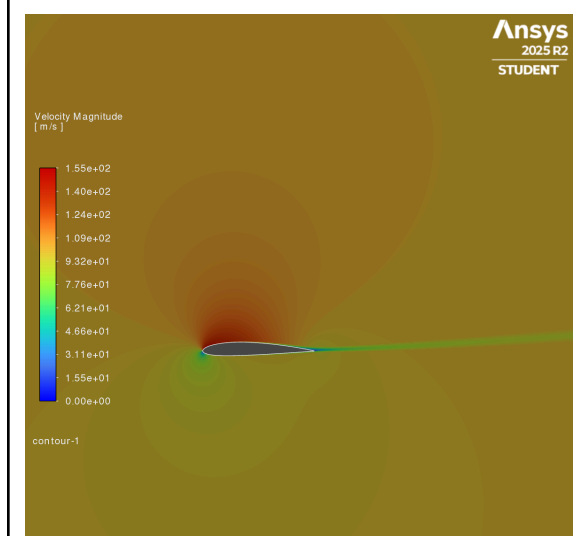
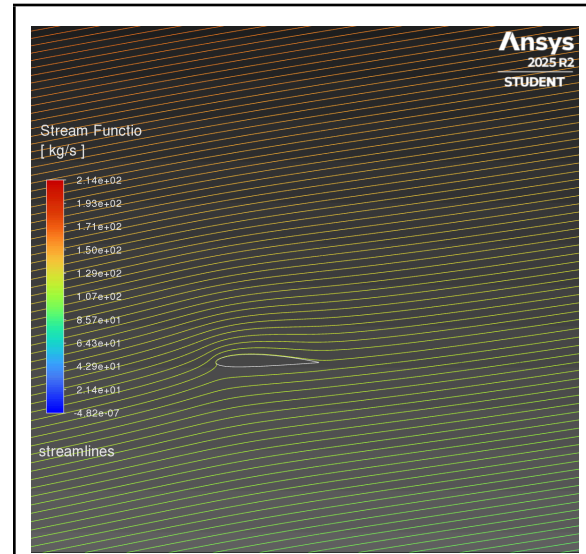
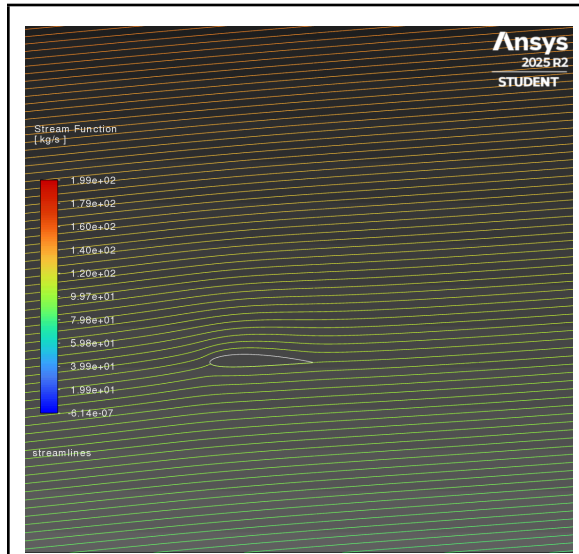
Computational Investigation of Airfoil Aerodynamics and Flow Control

Krishiv Patel

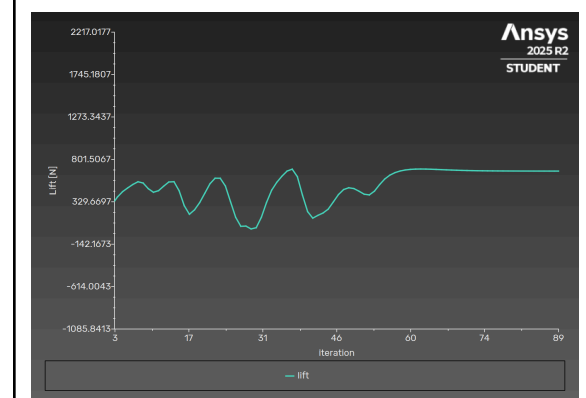
*Department of Mechanical and Aerospace
Engineering, Rutgers University, Piscataway, New
Jersey 08854*

Streamlines NACA 2412
at 0, 5, 10, 15, 20, 25
degrees AoA

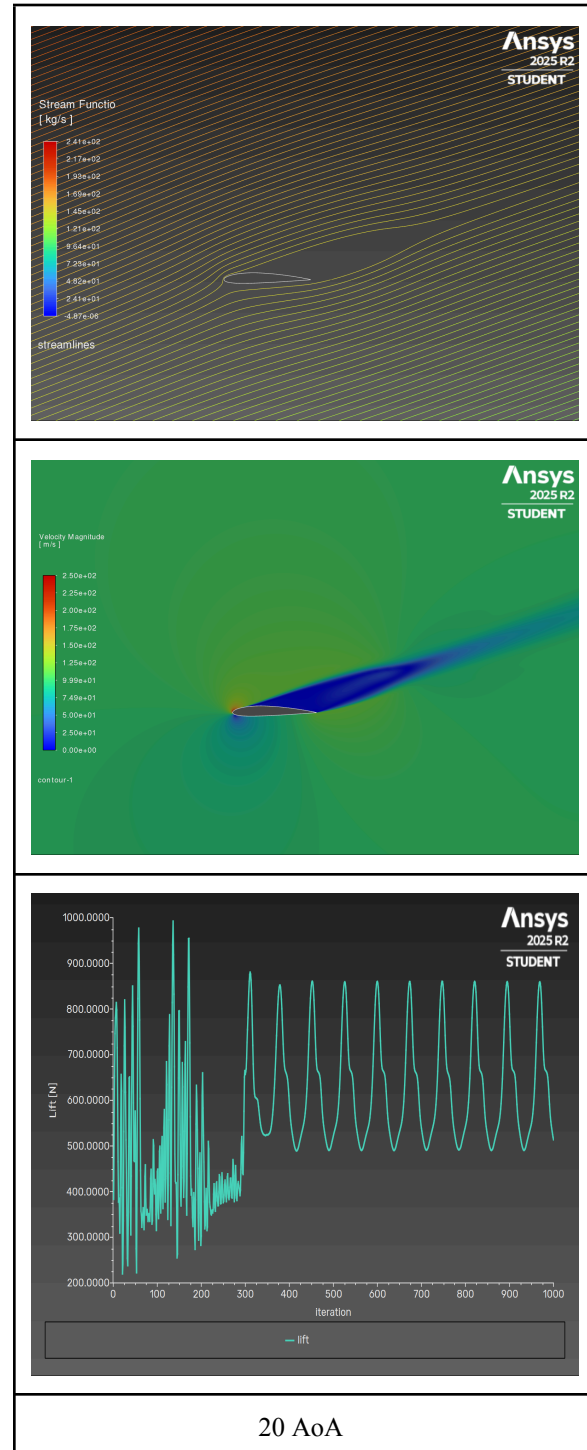
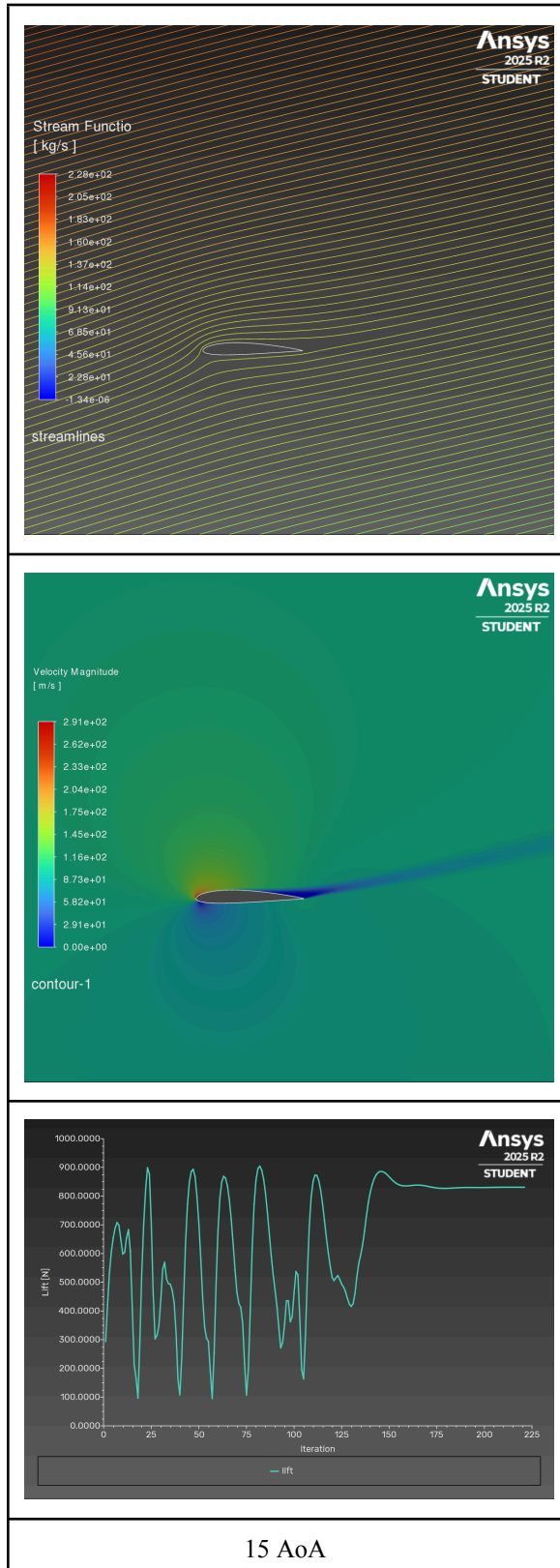


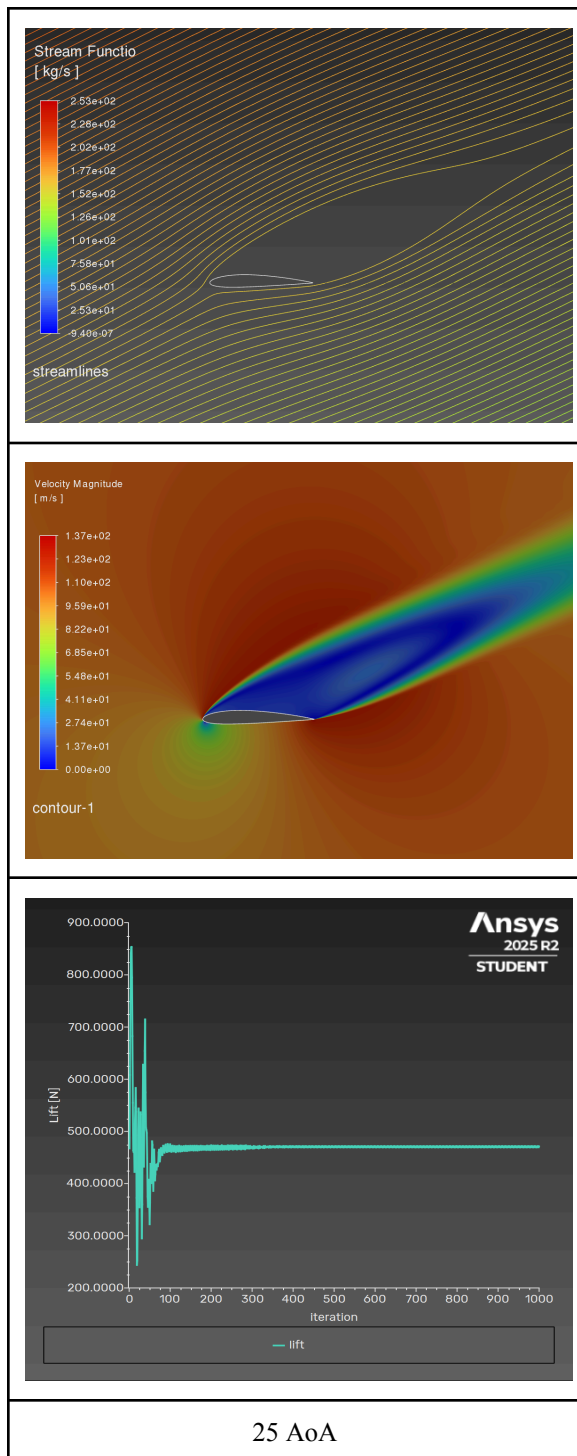


5 AoA

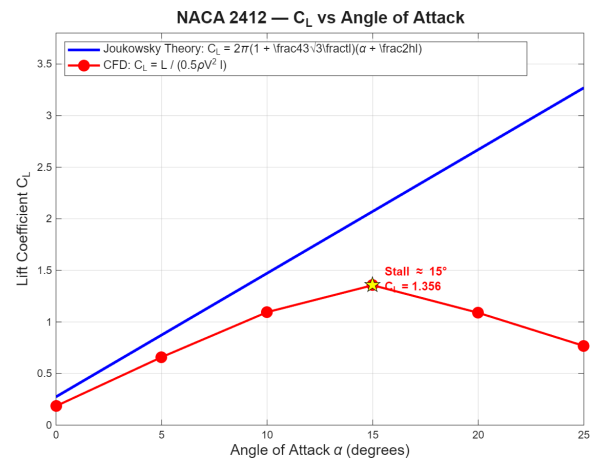


10 AoA





Plots of C_L using Experimental Values Vs. Joukowski



=== Standard Formula $CL = L / (0.5 \cdot \rho \cdot V^2 \cdot l)$ ===

AoA (deg)	Lift (N)	CL
0.0	113.1700	0.1848
5.0	403.6600	0.6590
10.0	670.8596	1.0953
15.0	830.6748	1.3562
20.0	667.1055	1.0892
25.0	470.5061	0.7682

=== Joukowski Theory ===

Thickness correction factor: 1.0924
Camber term ($2h/l$): 0.0400 rad

AoA (deg)	AoA (rad)	$CL_{Joukowski}$
0.0	0.0000	0.2745
5.0	0.0873	0.8735
10.0	0.1745	1.4725
15.0	0.2618	2.0714
20.0	0.3491	2.6704
25.0	0.4363	3.2694

Standard Formula Calculation:

$$C_L = L / (q_\infty \cdot c)$$

Where:

- $q_\infty = \frac{1}{2}\rho V^2 = \frac{1}{2} \times 1.225 \times 100^2 = 6125 \text{ Pa}$
- $c = 0.1 \text{ m}$
- So denominator = $6125 \times 0.1 = 612.5$

AoA (°)	Lift (N)	CL = L/612.5
0°	113.17	0.1848
5°	403.66	0.6596
10°	670.86	1.0952
15°	830.67	1.3562
20°	667.11	1.0891
25°	470.51	0.7682

Joukowsky Theory:

$$C_L = 2\pi \left(1 + \frac{4}{3\sqrt{3}} \frac{t}{l}\right) \left(\alpha + \frac{2h}{l}\right)$$

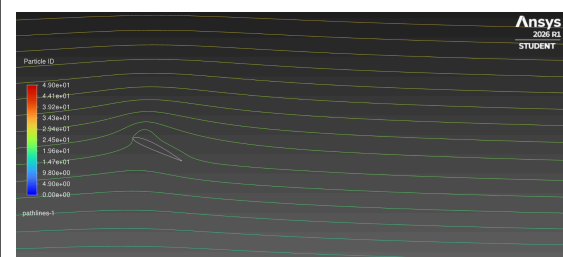
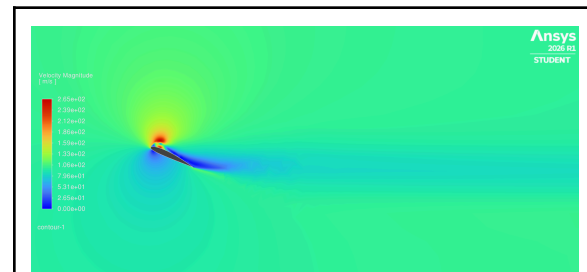
For NACA 2412:

- $t/l = \mathbf{0.12}$ (12% thickness)
- $h/l = \mathbf{0.02}$ (2% camber)
- α in **radians**

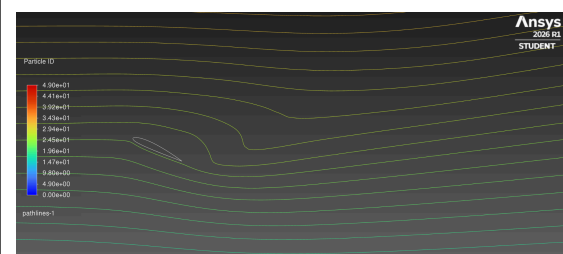
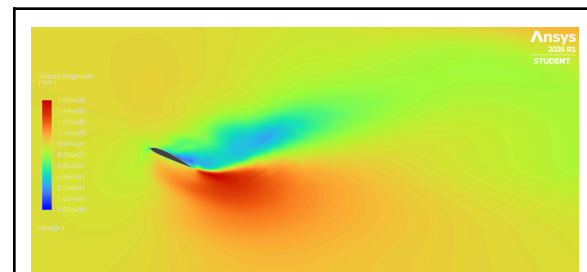
So: $CL = 2\pi(1 + 0.7698 \times 0.12)(\alpha + 0.04)$

AoA (°)	α (rad)	C_L Joukowsky
0°	0	0.2745
5°	0.0873	0.8731
10°	0.1745	1.4717
15°	0.2618	2.0703
20°	0.3491	2.6689
25°	0.4363	3.2675

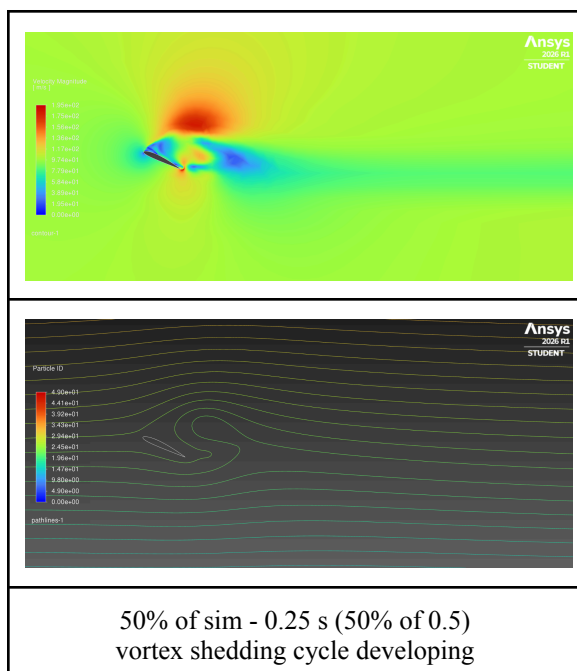
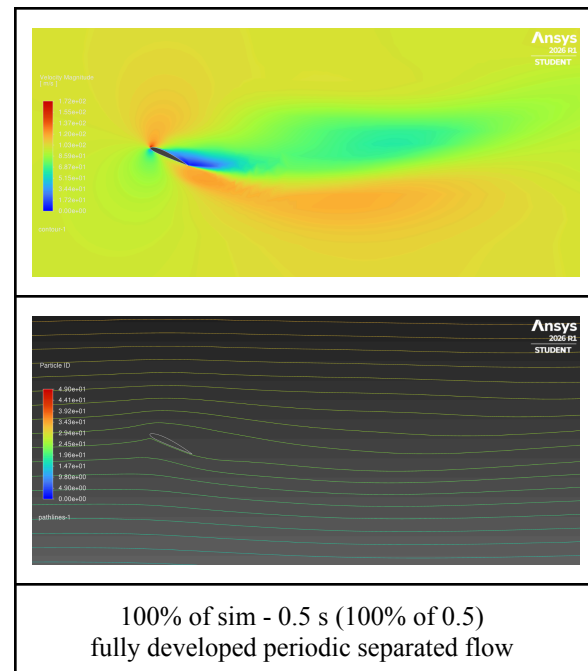
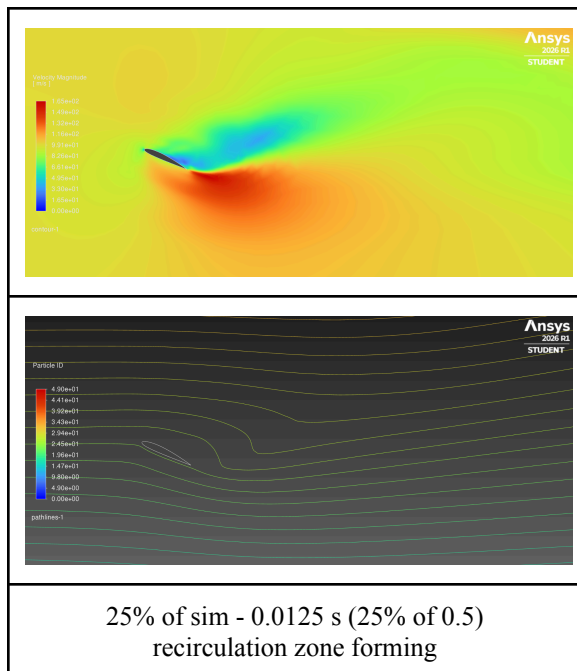
Plots of Streamlines and Velocity Contour AoA Unsteady



5% of sim - 0.025 s (5% of 0.5)
initial flow startup transient

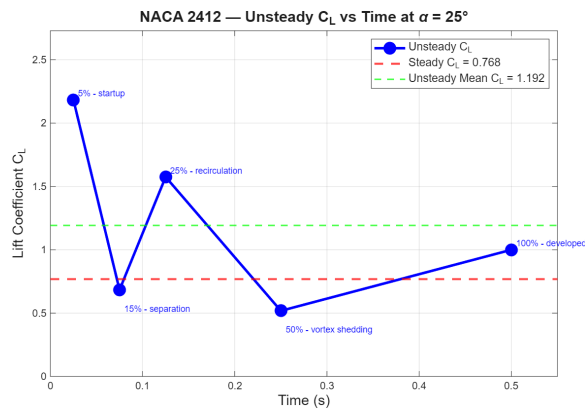


15% of sim - 0.075 s (15% of 0.5)
separation beginning to develop



Analysis:

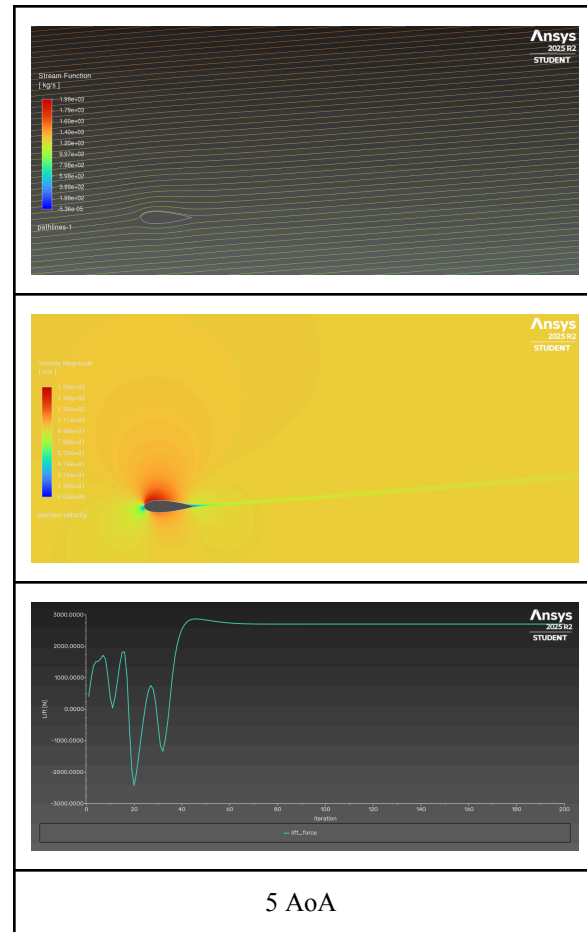
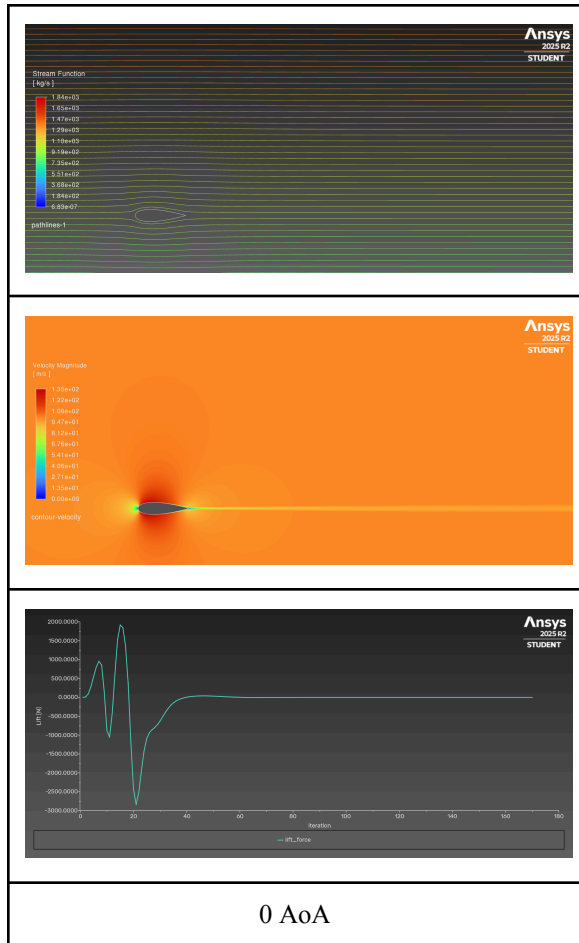
The unsteady simulation at 25° reveals the temporal evolution of flow separation that a steady solver cannot capture. In the early time steps, the flow undergoes an impulsive startup transient with rapid leading-edge separation. As the simulation progresses, a large recirculation zone develops in the wake with periodic vortex structures forming and shedding. By $t=0.5s$, the flow has reached a fully developed separated state, with the boundary layer completely detached from the upper surface across the entire chord length. The steady solver captured only the time-averaged end state, missing the dynamic formation process entirely. This demonstrates that under deep-stall conditions (25°), an unsteady formulation is physically necessary to accurately model the flow behavior, as the separation is inherently time-dependent and cannot be represented by a single converged steady solution.

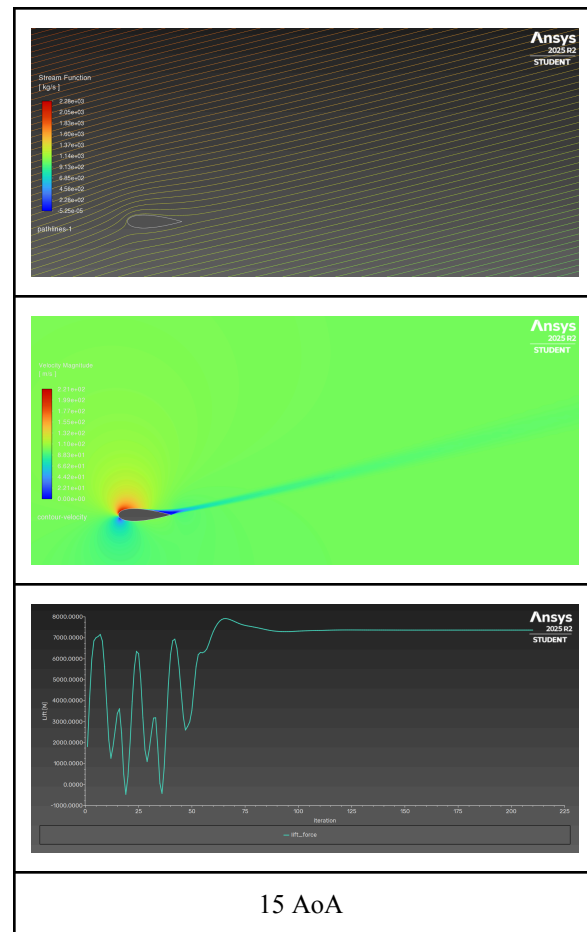
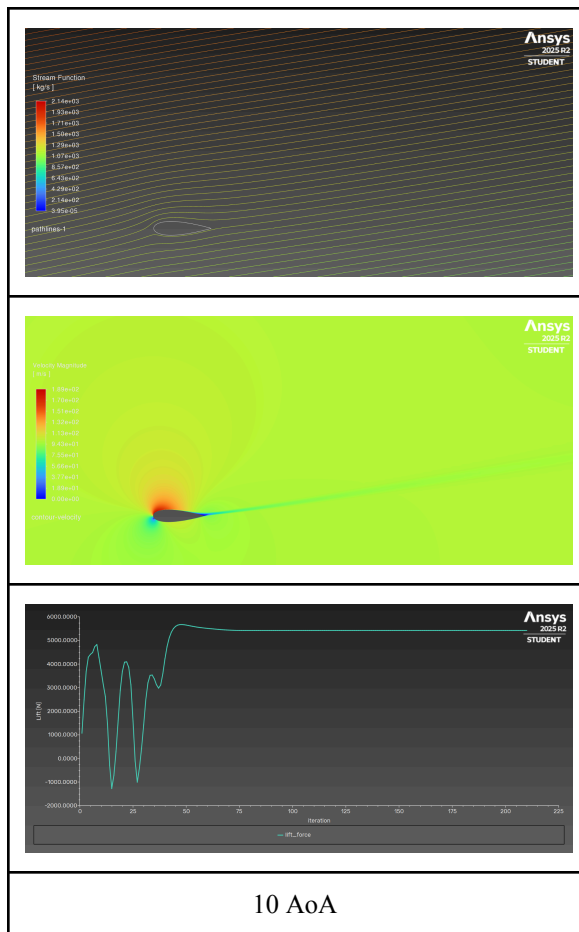


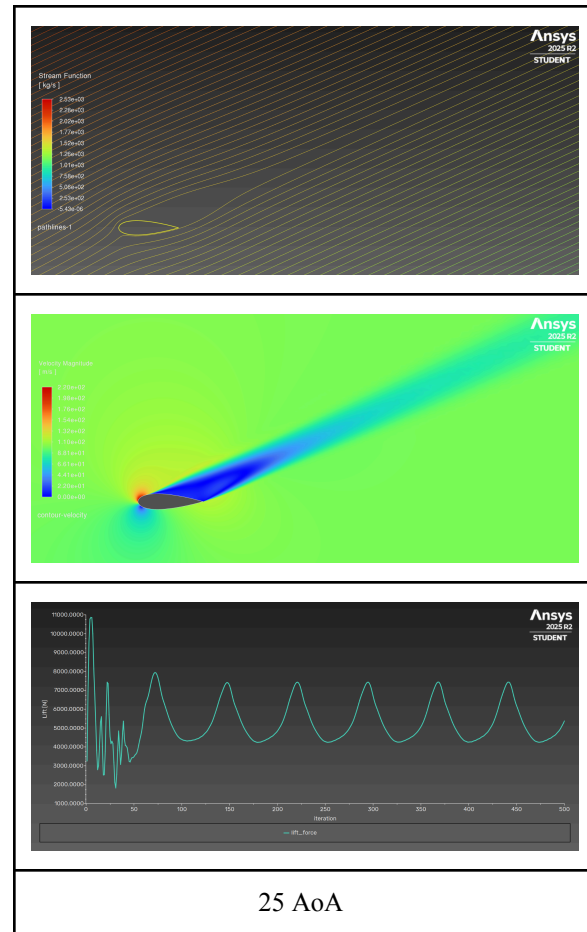
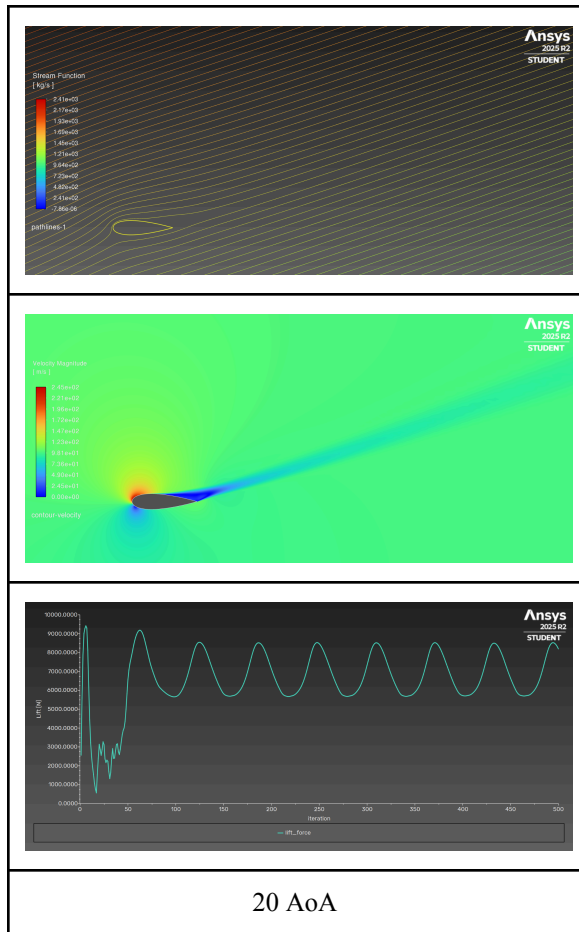
The C_L vs time plot quantitatively confirms the unsteady nature of the stalled flow. At startup ($t=0.025s$), the C_L spikes to 2.18, significantly above both the steady and mean values, where the airfoil initially generates excess lift before the wake fully develops. The C_L then drops sharply to 0.68 at $t=0.075s$ as the leading-edge separation fully establishes and the lift-generating pressure differential collapses. The subsequent oscillation between peaks (~ 1.57 at 25%, ~ 0.52 at 50%, ~ 1.00 at 100%) is the signature of periodic vortex shedding. Alternating vortices shed from the trailing edge cause cyclic pressure fluctuations that drive the lift up and down.

Notably, the unsteady time-averaged C_L of 1.192 is 55% higher than the steady solver value of 0.768. This discrepancy arises because the steady solver captures only the fully separated quasi-static state, while the unsteady simulation captures the dynamic lift contributions from the vortex structures that continuously reform and shed. This demonstrates that, for post-stall analysis, steady simulations underpredict the time-averaged lift, and an unsteady formulation is required to obtain physically accurate results.

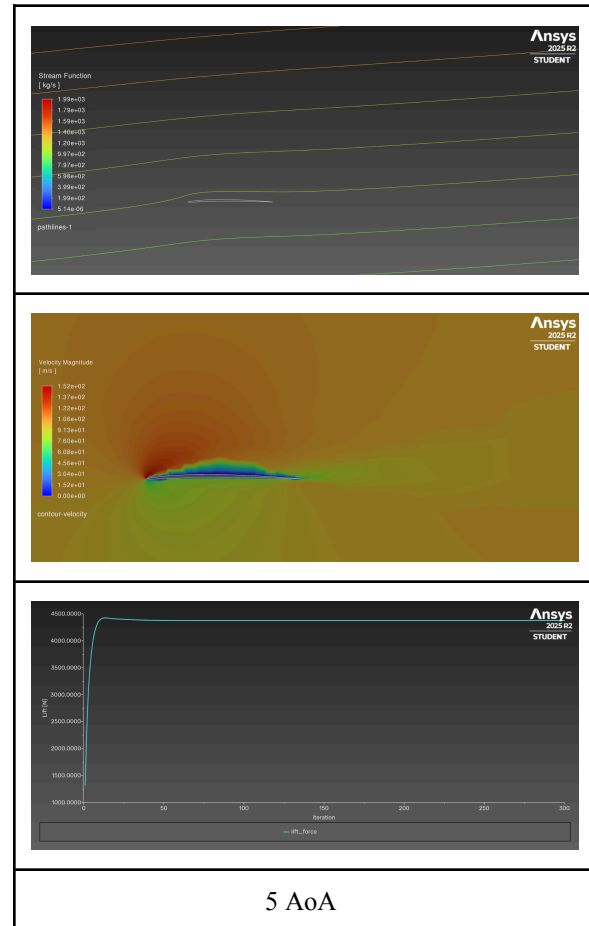
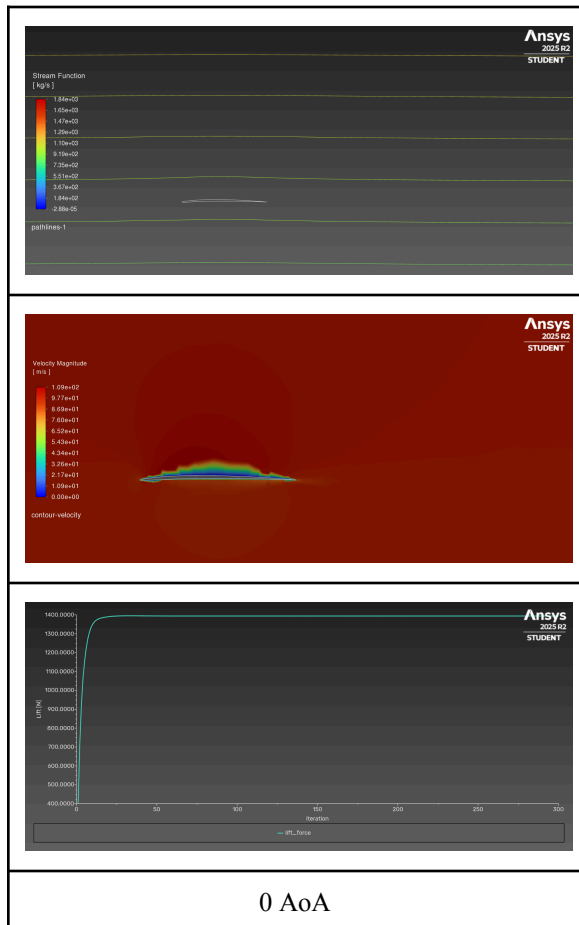
Plots of NACA 0024

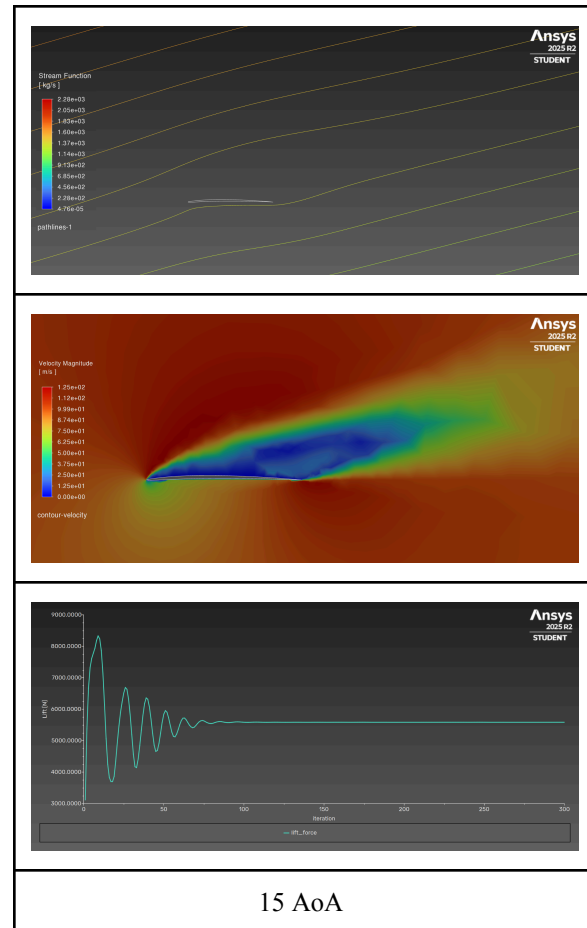
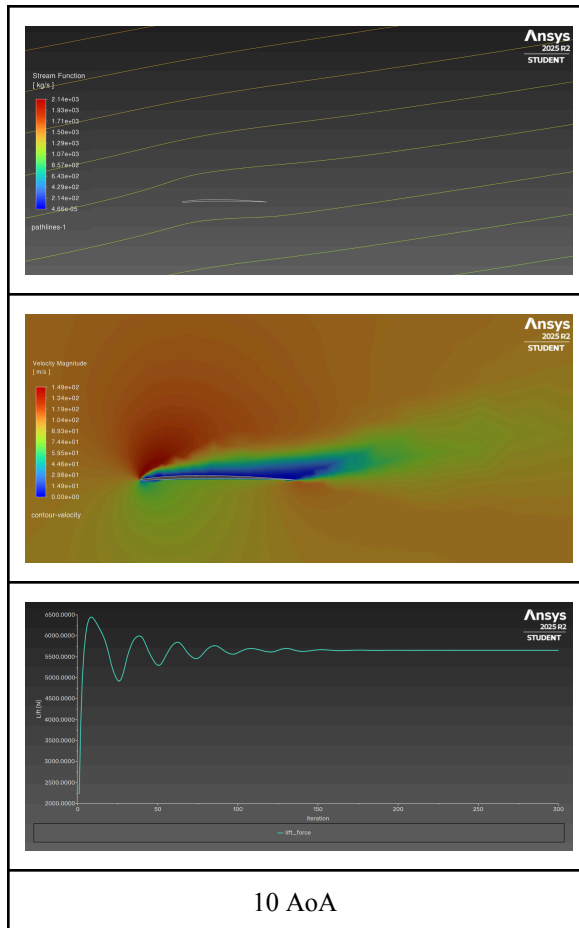


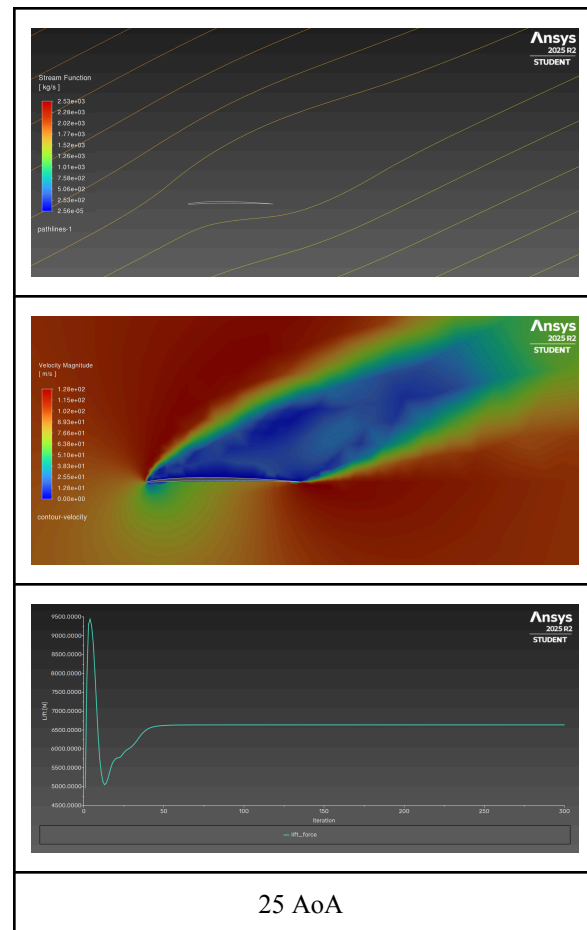
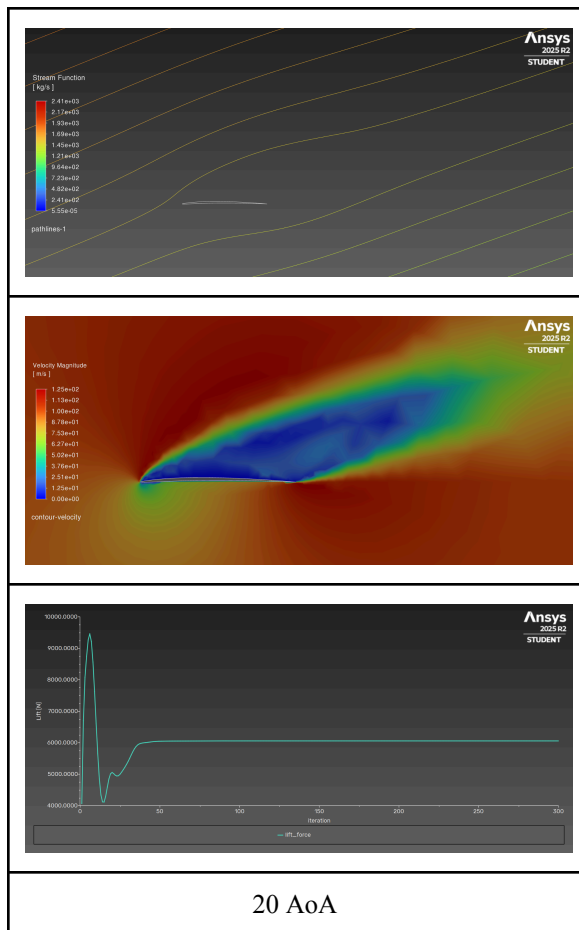




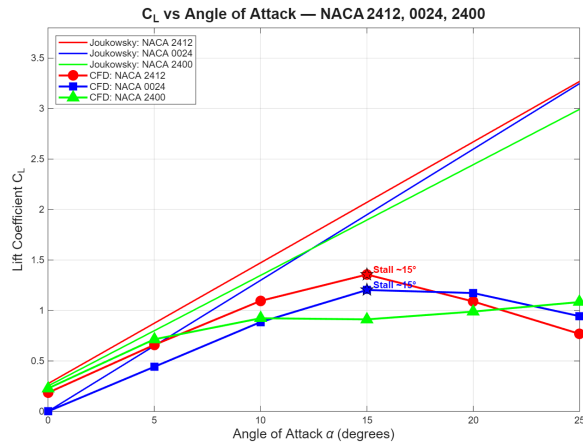
Plots of NACA 0024







Comparison of All Airfoils



NACA 2412 (camber + moderate thickness) stalls at around 15° , with a peak C_L of ~ 1.36 . The camber creates lift even at 0° and delays stall by keeping the flow attached longer on the upper surface.

NACA 0024 (symmetric, thick) stalls around 15° but with a lower peak C_L . Zero camber means $C_L = 0$ at 0° AoA. The extra thickness (24%) slightly delays separation compared to a thin symmetric airfoil.

NACA 2400 (cambered, near-zero thickness) shows irregular behavior — the non-monotonic dip at 15° is likely a **mesh artifact** due to the near-zero thickness, making it extremely difficult to mesh accurately. A near-zero-thickness airfoil is physically unrealistic and hard to simulate. In reality, a thicker cambered airfoil would stall more gradually.

General conclusion: Camber raises C_L at all angles and shifts the zero-lift angle below 0° . Thickness helps smooth the stall by delaying the leading-edge separation. An airfoil with both camber and sufficient thickness (such as the NACA 2412) achieves the best combination of high lift and a gradual stall onset.